

Week 8 Exercise.R* mydata2 mydata student_pass student_data

```

1 library(dplyr)
2 library(readr)
3 student_data<- read.csv("C:/UTP/yr 2 sem 2/datascience/Week 8 Exercise/student_data.csv")
4 View(student_data)
5
6 #1) student_pass
7 student_pass<- student_fail<- student_data[student_data$final_exam_mark< 80,]
8 View(student_pass)
9
10 #2) Ascending order based on course_work_mark
11 mydata<- student_data %>% filter(final_exam_mark > 80) %>% arrange(coursework_mark)
12 View(mydata)
13
14 #3) Only Student Name & Final Exam Mark
15 mydata2 <- student_data%>% select(student_name,final_exam_mark)
16 View(mydata2)
17
18 #4) Number of rows and columns
19 nrow(student_data)
20 ncol(student_data)
21
22 #5) Explain how to obtain details, show dataset details based on R query
23 str(student_data) #Structure of the dataset
24 summary(student_data) #Stats Summary
25 head(student_data) #First 6 Rows of the dataset
26 tail(student_data) #Last 6 Rows of the dataset
27 nrow(student_data) #Number of rows in the dataset
28 ncol(student_data) #Number of columns in the dataset
29
30 #Identify & Detect outliers in the dataset, explain and demonstrate using R
31 boxplot(student_data$final_exam_mark,
32        main = "Boxplot of Final Exam Marks",
33        ylab = "Marks",
34        col = "lightblue")
35
36 Q1 <- quantile(student_data$final_exam_mark, 0.25, na.rm = TRUE)
37 Q3 <- quantile(student_data$final_exam_mark, 0.75, na.rm = TRUE)
38 IQR_value <- IQR(student_data$final_exam_mark, na.rm = TRUE)
39
40 lower_extreme <- Q1 - 1.5 * IQR_value
41 upper_extreme <- Q3 + 1.5 * IQR_value
42 cat("Lower Extreme Boundary: ", lower_extreme)
43 cat("Upper Extreme Boundary: ", upper_extreme)
44
45 outliers <- student_data %>% filter(final_exam_mark < lower_extreme | final_exam_mark > upper_extreme)
46 cat("Identified Outlier Records: \n")
47
48 if(nrow(outliers) > 0) {
49   print(outliers)
50 } else {
51   cat("No outliers detected in the final_exam_mark column.\n")
52 }
53

```

Console Terminal Background Jobs

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R - R 4.0.0 - ~/Week 8 Exercise/
> mydata<- student_data %>% filter(final_exam_mark > 80) %>% arrange(coursework_mark)
> View(mydata)
[1] 50
> nrow(student_data)
[1] 5
> str(student_data) #Structure of the dataset
'data.frame': 50 obs. of 5 variables:
 $ student_id : chr "S001" "S002" "S003" "S004" ...
 $ student_name : chr "Elizabeth Miller" "William Garcia" "Sarah Johnson" "William Davis" ...
 $ gender : chr "Female" "Male" "Female" "Male" ...
 $ coursework_mark: int 40 59 79 56 74 41 64 73 70 67 ...
 $ final_exam_mark: int 41 63 64 62 56 35 84 60 64 65 ...
> summary(student_data) #Stats Summary
 student_id student_name gender coursework_mark final_exam_mark
Length :50 Length :50 Length :50 Min. :34.00 Min. :34.00
N.unique :50 N.unique :42 N.unique :2 1st Qu.:51.00 1st Qu.:51.50
N.blank :0 N.blank :0 N.blank :0 Median :62.00 Median :56.50
Min.nchar :4 Min.nchar:10 Min.nchar:4 Mean :62.42 Mean :57.88
Max.nchar :4 Max.nchar:17 Max.nchar:6 3rd Qu.:73.00 3rd Qu.:64.75
Max. :97.00 Max. :84.00
> head(student_data) #First 6 Rows of the dataset
 student_id student_name gender coursework_mark final_exam_mark
1 S001 Elizabeth Miller Female 40 41
2 S002 William Garcia Male 59 63
3 S003 Sarah Johnson Female 79 64
4 S004 William Davis Male 56 62
5 S005 John Smith Male 74 56
6 S006 Sarah Rodriguez Female 41 35
> tail(student_data) #Last 6 Rows of the dataset
 student_id student_name gender coursework_mark final_exam_mark
45 S045 Susan Brown Female 77 77
46 S046 Mary Davis Female 70 60
47 S047 Linda Rodriguez Female 50 54
48 S048 Jessica Garcia Female 63 54
49 S049 Linda Davis Female 61 53
50 S050 Robert Brown Male 73 57
> nrow(student_data) #Number of rows in the dataset
[1] 50
> ncol(student_data) #Number of columns in the dataset
[1] 5
> boxplot(student_data$final_exam_mark,
+ main = "Boxplot of Final Exam Marks",
+ ylab = "Marks",
+ col = "lightblue")
> Q1 <- quantile(student_data$final_exam_mark, 0.25, na.rm = TRUE)
> Q3 <- quantile(student_data$final_exam_mark, 0.75, na.rm = TRUE)
> IQR_value <- IQR(student_data$final_exam_mark, na.rm = TRUE)
> lower_extreme <- Q1 - 1.5 * IQR_value
> upper_extreme <- Q3 + 1.5 * IQR_value
> cat("Lower Extreme Boundary: ", lower_extreme)
Lower Extreme Boundary: 31.625
> cat("Upper Extreme Boundary: ", upper_extreme)
Upper Extreme Boundary: 84.625
> outliers <- student_data %>% filter(final_exam_mark < lower_extreme | final_exam_mark > upper_extreme)
> cat("Identified Outlier Records: \n")
Identified Outlier Records:
>
> if(nrow(outliers) > 0) {
+ print(outliers)
+ } else {
+ cat("No outliers detected in the final_exam_mark column.\n")
+ }
No outliers detected in the final_exam_mark column.
>

```

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Filter

	student_id	student_name	gender	coursework_mark	final_exam_mark
1	S007	David Jones	Male	64	84
2	S044	Jessica Brown	Female	77	84
3	S032	Joseph Smith	Male	94	82

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Filter

	student_id	student_name	gender	coursework_mark	final_exam_mark
7	S007	David Jones	Male	64	84
32	S032	Joseph Smith	Male	94	82
44	S044	Jessica Brown	Female	77	84

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Filter

	student_name	final_exam_mark
1	Elizabeth Miller	41
2	William Garcia	63
3	Sarah Johnson	64
4	William Davis	62
5	John Smith	56
6	Sarah Rodriguez	35
7	David Jones	84
8	Thomas Martinez	60
9	Michael Garcia	64
10	Jessica Garcia	65

